problem

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Assignment 1

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Water flows steadily through a fire hose and nozzle. The hose is 35 mm in diameter and the nozzle tip is 25 mm in diameter; water gauge pressure in the hose is 510 kPa. The exit speed and pressure are 32 m/s and atmospheric, respectively. Find the force at the nozzle–hose coupling and its sense.

$\boxed{F = 2{,}8\,\mathrm{kN} \quad \text{(tension)}}$ RESULT

image

Determine the gauge pressure at point a if liquid A has $\mathrm{SG} = 1{,}20$ and liquid B has $\mathrm{SG} = 0{,}75$. The liquid around a is water; left tank open to atmosphere.

$\boxed{p_a = \dots}$ RESULT

image